



ORIGINAL RESEARCH

PBNet: A Pixel-Wise Boundary Information Refinement Network for Road Scene Segmentation

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Abstract

Efficient and accurate semantic segmentation is particularly important in scene parsing. Despite the significant progress made by deep convolutional neural networks in semantic segmentation, lightweight models still struggle to achieve a balance between real-time performance and accurate boundary segmentation, particularly for complex and small objects in road scenes. To address these challenges, a lightweight pixel-wise boundary information refinement network is proposed for road scene segmentation. First, a multi-scale boundary information extraction module (MBIE) is proposed for deep pixel feature refinement. Second, a low-level boundary information extraction module (LBIE) is proposed for shallow feature refinement extraction. Finally, a feature reuse fusion module (FRF) is designed for boundary pixel information fusion. The MBIE module introduces a channel separation pyramid and cross-scale self-attention to capture long-range dependencies. The LBIE module uses dilated self-attention to enhance low-level feature semantics. The FRF module employs dual-attention mechanisms to fuse semantic and spatial information effectively. Experimental results on CamVid and Cityscapes datasets show that the proposed method achieves a mIoU of 74.61% and 69.33%, respectively, outperforming state-of-the-art scene segmentation methods in terms of segmentation accuracy while maintaining real-time performance.

Keywords Multi-scale boundary information refinement · Channel separation pyramid · Cross-scale self-attention · Feature reuse fusion · Road scene segmentation

Introduction

As a widely used technique in computer vision, semantic segmentation is used in the fields of autonomous driving, video surveillance, robot sensing, etc. Semantic segmentation is more complex than classification tasks because it considers the accuracy of both semantic and spatial

information simultaneously. With the increasing demand of real-time and large-scale scene for different applications, how to maintain efficient reasoning speed and high accuracy in high-resolution images has become an important research topic.

To improve the accuracy of semantic segmentation, the related research has improved based on image classification network, which was applied to semantic segmentation [1]. Current semantic segmentation networks are usually based on deep CNNs (such as Alexnet, VGG, GoogLeNet, ResNet and Xception) [2–6], and adopt encoder-decoder structure. The encoder is responsible for extracting the feature representations of the input image, and the decoder utilizes the feature representations to generate pixel-level segmentation. These models use deeper network structure to enlarge the receptive field, and extracts more abundant features of semantic information [7]. However, the output resolution of these networks is low, which leads to the degradation of the accuracy of spatial information.

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To solve this problem, researchers proposed three kinds of methods to ensure the accuracy of spatial information: canceling the steps of the last blocks in the encoder to improve the output resolution [8], adding multi-column parallel spatial branches [9], and fusing the lower-level feature maps in the encoder-decoder based methods [10]. However, these methods have corresponding disadvantages. The first will lead to insufficient feature semantic information. When pyramid-like feature combination (e.g. pyramid pooling of atrous space) is added to fuse multi-scale context information to enrich features, the parameters will be multiplied. By adding multi-column parallel branches, the second can recover the spatial information by fusing the extra lightweight spatial branches without changing the backbone network. However, due to the shallow depth of the network, these parallel branches will introduce incorrect semantic information, which will lead to the degradation of the performance. As the current mainstream, the third type of methods concatenate the feature maps of each layer of the encoder with the output feature maps of the decoder, which effectively combine the spatial information of the encoder with the semantic information of the decoder. In U-Net [11], although the influence of spatial information in the output features of the encoder on the network performance is considered, the fusion operations of the high-resolution feature maps bring massive additional calculations. Moreover, it leads to lower fusion efficiency and information redundancy without corresponding feature extraction for each layer of the encoder output features. Most of recent high-precision semantic segmentation models adopt large-scale feature extraction module as the backbone, such as ResNet-101, and design complex feature processing module for segmentation tasks. They require large amount of storage and GPU resources, which are not suitable for real-time scenarios [12].

With the wide application of deep learning in autonomous driving industry, models are required to be deployed on edge computing devices for real-time segmentation. Lightweight real-time semantic segmentation model has attracted much attention and becomes a hot research field. Compared with the high precision semantic segmentation, the real-time semantic segmentation algorithm has the advantages of low delay, less resource consumption, and higher practicability etc., which tries to find the best balance between accuracy and computing complexity [13]. Recently, real-time semantic segmentation is improved and has achieved satisfactory performance in various benchmark tests [14]. Some studies have reduced computational complexity by limiting the size of input images, or by pruning redundant channels in networks, such as ENet [15], MobileNet [16], ShuffleNet [17], EfficientNet [18], and other lightweight networks. By designing more compact network structure, discrete

convolution and depthwise separable convolution are used to reduce parameters and computation to realize real-time segmentation on edge computing devices [19]. Although these methods are effective in efficiency, the shallow network will weaken the ability of feature discrimination, and easily lose the boundary and the spatial details around small objects, which brings great security risks to applications such as autopilot. To address this problem, a multi-branch framework that combines spatial details with contextual information was proposed [20]. In BiSeNetv1 [21], a shallow and wide sub-branch network is designed to capture spatial information, in which two parallel networks are used to process and fuse features of different scales. The ICNet [22] combines spatial details and contextual information by cascading subnetworks of different resolutions. However, the independence of sub-networks limits the learning ability and generalization ability of the model. On the other hand, some other methods usually input the image into the single-path system, and then enrich the information of the output features through the feature extraction module. For example, FastSCNN [23] uses the feature pyramid pooling module of PSPNet for information extraction of the output features of backbone. However, although the features of the original image are fused, the high-level context still lacks the combination with the previous features, and it is difficult to retain the spatial details and semantic information in the path at the same time. To enhance the learning ability and increase the receptive field, feature reuse is important, which urges researchers to find a lightweight method to incorporate multi-layer context into features. Inspired by this, a dual-channel and feature aggregation module is designed in DFANet to provide better context modeling and feature representation capabilities [24]. Inspired by the multi-scale attention mechanism in HRNet [25, 26], high-resolution feature maps are constructed at different resolutions in Lite-HRNet, which achieves satisfactory segmentation accuracy and real-time performance. The boundary region is a transition region between different classes, which provides clear edge information between objects and is essential for the algorithm to segment different semantic objects more accurately. Moreover, the boundary information also helps to keep the integrity of the semantic object and prevent the unnatural break in the segmentation result. By making full use of the boundary information, the semantic segmentation algorithm can understand the spatial distribution and structure relationship of different semantic categories more comprehensively and accurately.

However, the current semantic segmentation techniques still face challenge in dealing with boundary information. For example, the frequent pooling and up-sampling operations in U-Net and the derivative methods result in boundary information loss and further lead to missing or blurry

edges of the objects in the segmentation results. In DeepLab, atrous convolutions are used to increase the receptive field. However, but for some regions in fine categories, atrous convolutions are still not sufficient to capture boundary details especially for categories that with fine geometric features. To process multi-scale information efficiently, multi-resolution shared features are fused to expand the receptive field in SwiftNet, it cannot reach satisfactory segmentation boundary for objects such as street lights, billboards, etc [27].

To solve the problem, this paper proposes a lightweight pixel-wise boundary information refinement network (PBI-Net) using an encoder-decoder architecture for road scene semantic segmentation. In this work, the boundary information is refined to label the pixel categories of these regions more accurately, thus improving the accuracy of semantic segmentation. Two strategies are deployed, namely extracting further information from feature map of each layer of the encoder and designing effective information fusion method for decoder, to extract the features of high-level and low-level feature maps more effectively, and improve information capture ability. Firstly, a multi-scale boundary information extraction (MBIE) module is designed, which consists of a channel separation pyramid and self-attention. Multi-scale representations are extracted from convolution layers with different receptive fields and cross-scale self-attention structures. Secondly, the low-level boundary feature extraction (LBIE) module is designed for other low-level outputs of the encoder, which balances the spatial weight and semantic weight of the feature map with LBIE module. Thirdly, after refining the output features of the encoder, the feature reuse fusion (FRF) module is used in our redesigned decoder to realize the cross-layer feature aggregation of the high and low-level outputs after further feature extraction. The high-level feature map is guided by the low-level feature map with accurate spatial information to recover the boundary information of the high-level feature map. And the fused feature map with accurate semantic and spatial information is obtained to solve the problem of object boundary blurring and small target segmentation in scene. Finally, a hybrid loss function is designed to evaluate the difference between the predicted result and the ground truth. The proposed PBI-Net is validated on the road scene datasets CamVid [28] and Cityscapes [29], and achieves satisfactory segmentation performance compared with the state-of-the-arts.

The main contributions of this work are as follows.

- (1) The MBIE module is proposed based on channel separation pyramid and self-attention. The long-distance information dependence of feature map is captured by cross-scale self-attention module, and the accurate

dense features are extracted for pixel segmentation. The LBIE module is designed to balance the spatial weight and semantic weight of the feature map.

- (2) We propose the FRF module to realize cross-layer feature aggregation. The mapping information extracted from the features are weighted and fused one layer at a time. Therefore, the high-level semantic information and low-level spatial information are taken into account in mining the multi-scale context features.
- (3) The hybrid loss function based on CE Loss and Dice Loss is designed to balance the accuracy of pixel-wise segmentation and region-level consistency to address the class imbalance.
- (4) The pixel-wise boundary information refinement network is proposed for end-to-end semantic segmentation of road scenes. Comparative experiments on the CamVid and Cityscape datasets verify the effectiveness of the proposed method.

Related Work

Semantic Segmentation

Convolutional neural networks have proven effective in solving a variety of computer vision problems [30]. They can be divided into region growing segmentation and full convolutional network segmentation.

Region growing segmentation usually includes two main steps: candidate region generation and region classification. First, the candidate regions are generated by traditional image segmentation algorithms (e.g. edge detection and super-pixel segmentation). Second, feature extraction and classification algorithms are used to determine which semantic category each candidate region belongs to. Region growing methods rely on the generation and classification algorithms of candidate regions, in which the classification process can use the traditional machine learning method or deep learning method. For example, a deep learning model can be used for edge detection or hyper pixel segmentation in the candidate region generation. In the region classification stage, deep learning models, such as convolutional neural network (CNN) and recurrent neural networks (RNN), can be used to extract the features of candidate regions and classify them.

In 2015, Long et al. proposed a pixel-wise segmentation network FCN [31] based on encoder-decoder architecture, which pioneered a semantic segmentation method based on full convolutional networks. Compared with traditional region based segmentation, it can achieve pixel-wise classification, and the segmentation mask is generated by fully convolutional network without artificial features or explicit

image segmentation steps. By replacing the full-connection layer with convolution layer, the fully convolutional network can effectively process input images of any size while preserving spatial information. Although the FCN-based methods made progress in the field of semantic segmentation, their architecture leads to the loss or fuzziness of spatial information details, the imbalanced number of pixels in different categories. To address these limitations, subsequent research efforts have concentrated on leveraging global or multi-scale context information to enhance the semantic richness of feature representations. Based on the encoder-decoder architecture, PSPNet, Deeplab [32], DenseASPP [33], DANet [34] etc. have been proposed. Pyramid pooling modules are used in PSPNet to efficiently extract context information across dimensions. In DeeplabV2 [35] and DeeplabV3 + [36], a spatial pyramid pooling constructed by parallel dilated convolution with variable dilated rate to integrate context information. In DANet and EMANet [37], different decoder architectures are designed based on self-attention for information fusion. These methods received satisfactory performance on context information. However, their performance in terms of multi-scale context fusion efficiency needs further improvement.

Real-Time Semantic Segmentation

The goal of real-time semantic segmentation algorithm is to generate high-quality predictions under limited computational load. Semantic segmentation usually implies the pre-trained backbone for image classification, such as ResNet, VGG, Xception, DenseNet, etc. However, these backbones can hardly meet the real-time requirements. Lightweight backbones are necessary to achieve more efficient and effective segmentation performance with fewer parameters. The semantic segmentation network requires the model to be able to output features with strong semantic information while not losing resolution information. Therefore, researchers focus on the real-time segmentation model which can extract strong semantic information and integrate resolution information. In SegNet, pooling indices are used to design a symmetric encoder-decoder architecture that greatly reduces network parameters [38] because of unpooling. ENet considers reducing the number of down samplings in pursuit of a more compact framework. Because it omits the final stage of the model, the model's receptive field is too small to correctly segment larger objects. In ESPNet, a spatial pyramid module is used to improve computing efficiency [39]. ICNet uses multi-scale images as input and uses cascading networks to improve efficiency. BiSeNet addresses the problems of spatial information loss and receptive field contraction in existing lightweight semantic segmentation networks by introducing the spatial

and context paths while reducing the computation [21]. In ICNet and BiSeNet, only one branch uses deep CNN for feature extraction while the other parallel branches are used to compensate for resolution details, which are shallow in depth. However, this architecture may yield incorrect semantic information and result in negative effect on the segmentation result. In addition to the network structure, researchers also aim to improve model learning ability by designing appropriate loss function. For example, SwiftNet can detect the pixels near the boundary by means of boundary sensing loss, which increases the ability to perceive the boundary information [40]. However, the fusion operation of its decoder cannot eliminate redundant information, and it may not be able to segment the detailed boundary features accurately in complex scenes.

Inspired by the aforementioned methods, an efficient and lightweight decoder is designed in this work without changing the backbone. In the proposed PBINet, the lightweight pre-trained MobileNetV2 is used as the backbone to ensure the overall lightweight and extendibility of the network. Moreover, the proposed MBIE and LBIE modules can be easily embedded into other networks to improve network performance.

Refinement of Boundary Information

In light-weight semantic segmentation, the refinement of boundary information can locate the segmentation result more accurately, make the boundary between different semantic categories more accurate, reduce fuzzy boundary, and reduce erroneous segmentation [41]. Therefore, boundary information refinement plays a important role in lightweight semantic segmentation. The existing decoder networks recover the boundary and spatial information by integrating the low-level features into the high-level features. However, the direct fusion of low-level and high-level feature maps affects segmentation quality. To address the problem, researchers designed various structures based on encoder-decoder architecture to refine and optimize segmentation results. In LiteSeg, boundary information is restored by improving ASPP structure, dilated convolution with different dilation rates [42]. In DFANet, multiple interconnected encoding streams are designed, which incorporate high-level context into the encoding features and refine the high-level features with slightly increased parameters. High-resolution branches are designed in CABiNet to recover spatial details [43] and long distance and local context dependencies are captured by using globally and locally aggregated context branches. In CSRNet, feature maps are refined progressively by a multiple-stage segmentation network [44]. These methods either focus on building object's internal consistency by aggregating global or multi-scale

context information, or simply combining semantic features with boundary features to refine object details. However, they ignore the long distance dependencies between the internal objects and their boundaries.

Considering the effect of information dependence on boundary information refinement, this paper improves the feature representation ability in feature space, guide the context information aggregation by boundary information, and the information flow between different stages is guaranteed by residual connection. First, the low-level boundary information is used to focus on boundary area and extract high-level semantic features. Secondly, effective feature fusion is achieved by aggregating context boundary information to enhance intra class consistency.

Attention Mechanism

In the structure of encoder-decoder based on FCN, the decoder is responsible for remapping the features obtained by the encoder to the original image. Due to the influence of spatial information and channel information, there are low-level features and high-level features from the encoder in different levels of the decoder. To make full use of these features, the application of attention mechanism is particularly important, which can be used to learn the relationship between features and adjust their weight ratio according to their importance. The attention mechanism can be divided into soft attention and self-attention. Soft attention obtains global context information by calculating the association weights between each pixel and all the other pixels, selecting fine-grained important pixels, and then weighting each pixel for aggregation. Based on soft attention, channel attention is introduced in SENet to weight and redistribute channel information [45]. In CBAM, serial channel attention and spatial attention modules [46] are proposed by considering the ideas of maximization pooling. Self-attention captures the long-distance dependence of features by calculating the association weights between each pixel and its neighbors. DANet and DSANet [47] adopt two self-attention modules, which are composed of spatial attention and channel attention, to aggregate the global dependence of local features on spatial and channel dimensions, and help to predict the pixel-wise labels of each stage. By virtue of AEFM module in BFMNet, attention mechanism is used to enhance and fuse bilateral feature [48]. However, these attention mechanisms are not sensitive enough to spatial variations to fully take into account changes in the shape and structure of objects. This would result in some marginal areas of information that is easy to ignore especially for the imbalanced categories.

In this work, soft attention and self-attention are combined to balance the use of global and local information, and

to further improve the performance of semantic segmentation. The attention mechanism is introduced to construct the feature reuse module of low-level features and high-level features. The module adaptively selects and adjusts the contributions of low-level and high-level features to better integrate information. Furthermore, low-level and high-level features can be fused in different levels of the decoder through the attention mechanism, which makes full use of different levels of semantic information. The module can help to improve the precision of semantic segmentation and boundary details, and make the network better adapt to different scales and complexity of the image scene.

Methodology

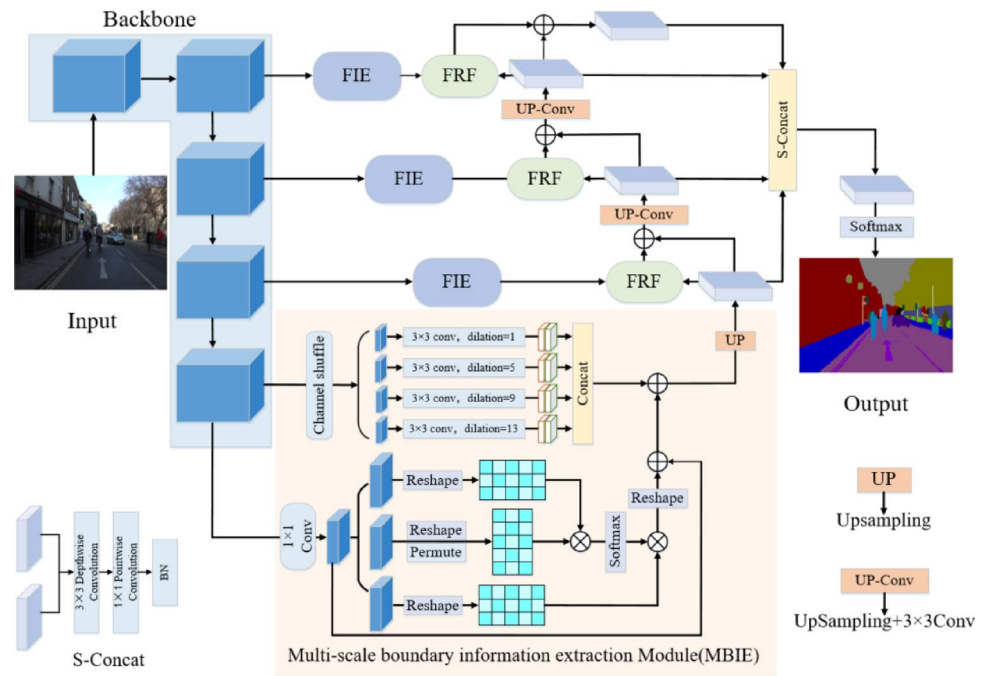
In this section, the PBINet network is described as follows. Section 3.1 introduces the overall network architecture. Section 3.2, 3.3 and 3.4 introduce the proposed MBIE module, LBIE module, and FRF module, respectively. Finally, the mixed loss function used in this work is presented in Sect. 3.5.

Network Architecture

Since ENet and ICNet were proposed, the lightweight real-time semantic segmentation network has received extensive attention. Recently, the lightweight networks BiSeNetv3 and SFRSeg achieve satisfactory results in the field of semantic segmentation. However, they both adopt well-designed backbone network pre-trained on ImageNet, which is not easy to expand. In this work, we explore and design a novel decoder network and plug-and-play three modules (MBIE, LBIE, FRF) to realize the extendable real-time road scene semantic segmentation. The overall architecture of the proposed PBINet is shown in Fig. 1. The lightweight backbone MobileNetV2 is used as the encoder, in which the last layer of down-sampling operation is canceled because of the low channel dimension. The MBIE, LBIE and FRF modules proposed in this paper are introduced in the decoder part.

For semantic segmentation tasks, especially for road scene segmentation that used for autonomous driving, the features of all the resolutions are helpful for segmentation [49]. Low-resolution semantic features are helpful for image region classification, while high-resolution features are helpful for generating clear and detailed boundaries for high-resolution prediction. The two kinds of context information are fused to restore the image resolution, and perform satisfactorily pixel-level prediction. In the decoder, the multi-stage feature fusion is guided by the deep context information, and the dense prediction is realized by deep iteration over long distances and residual connection.

Fig. 1 The network architecture of the proposed PBINet



First, the original image $F \in R^{3 \times H \times W}$ is fed into the backbone MobileNetV2, and the different feature maps in each stage are generated, among which F_1 , F_2 , F_3 and F_4 are the maps of the last four stages, respectively. The feature F_1^* which is rich in boundary information is extracted through the MBIE module from F_1 of the encoder. The other three maps are processed through the LBIE module to perform feature weighting. Second, the residual connection network and FRF module introduced in this work are used to realize information fusion layer by layer to enhance the ability of detail feature representation. The up-sampling and 1×1 convolution are used to recover feature dimensions. Residual connection is used to form a more efficient decoder to deal with the information degradation problem in each layer of the encoder. Then, context aggregation is guided by depth-wise separable convolution to generate feature map F_o . Finally, the aggregated feature map F_o is fed into the pixel-wise predictor (Softmax) to obtain the final semantic segmentation result. The mixed loss function based on crossentropy loss and binary crossentropy loss is used to enhance the feature learning ability of the model.

Multi-Scale Boundary Information Extraction Module

The atrous spatial pyramid pooling (ASPP) module was proposed in DeepLab network, in which multiple convolution with different dilated rates is used in a parallel convolution structure. Because the receptive field of dilated convolution

varies with the sampling rate, the ASPP module can accurately and effectively recognize multi-scale features. However, it can inevitably lead to the loss of local details due to the grid effect. Moreover, the convolution of parallel four high dimensions results in the sharply raised parameters. Although pyramidal features can significantly improve network performance, it is limited by the long-distance dependence of information and the scale of small target features. To further refine the boundary information of the output features of the encoder and address the problems of the model efficiency degradation caused by the unbalanced contribution of the features and the large number of parameters, a MBIE module which consists of a channel separation pyramid and self-attention is designed as shown in Fig. 1.

Channel shuffle is used to encourage cross-group information exchange among the grouped convolutions, which helps to mitigate the representational bottleneck caused by channel splitting. This enhances feature diversity and improves the model's ability to capture multi-scale boundary information without increasing computational cost. To build the channel separation pyramid, the high-level feature $F_1 \in R^{320 \times h \times w}$ (h , w are width and height, respectively) with 320 channels which generated by the backbone MobileNetV2 is fed into the module, and the channels of F_1 are divided into four groups by channel random mixing operation to get $F_i^1 \in R^{80 \times h \times w}$, $i \in [1, 2, 3, 4]$

$$F_i^1 = \text{channel_shuffle}(F_1) \quad (1)$$

Then, four sets of dilated convolutions with different dilation rates are used to expand the receptive field without changing the resolution of the feature map

$$y[i] = \sum_{k=1}^K F_i^1[i + r \cdot k] \cdot w[k] \quad (2)$$

where r is the dilation rate, $w[k]$ is the k parameter of the filter, and K is the size of convolution kernel. The four branches are fused using concat operation to get F_1^σ . The dilation rates (1, 5, 9, 13) were chosen to capture multi-scale context information while avoiding the grid effect commonly associated with larger dilation rates. These values were empirically validated through ablation studies (see Table 2) to provide a balance between receptive field size and feature resolution.

For the self-attention, $F_s \in R^{128 \times h \times w}$ is obtained by 1×1 convolution, it is then divided into three identical branches F_s^1 , F_s^2 and F_s^3 . The height and width of the feature map F_s^1 are mapped to one dimension by reshape operation, and the global information F_s^σ is obtained as follows by the inner product of F_s^1 and the processed F_s^2 by reshape operation

$$F_s^2 = \text{Res}(F_s^1) * \text{Per}[\text{Res}(F_s^2)] \quad (3)$$

where $F_s^2 \in R^{128 \times 128}$, $*$ refers to the inner product of the matrix, $\text{Res}()$ refers to the reshape operation, and $\text{Per}()$ refers to the transpose operation. F_s^2 is activated by Softmax to extract the channel weights, which is multiplied by F_s^3 after reshape to extract the context information $F_s^\delta \in R^{128 \times h \times w}$

$$F_s^\delta = \text{Softmax}(F_s^\sigma) * \text{Res}(F_s^3) \quad (4)$$

Then F_s^δ is restored to its original dimension and fused with F_s in the form of residual connection to get the final output $F_s^\sigma \in R^{128 \times h \times w}$ of the attention part

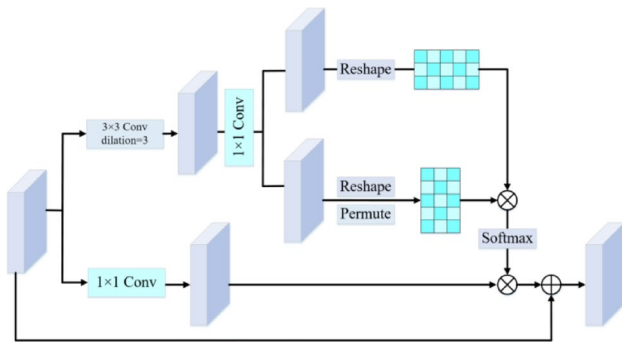


Fig. 2 The architecture of the LBIE module

$$F_s^\sigma = F_s + \text{Res}(F_s^\delta) \quad (5)$$

Finally, the outputs of the channel separation pyramid and self-attention are added and up-sampled to output F_M of MBIE module.

The module can sense a wider range of context information by using dilated convolution and self-attention, and adaptively weight the features of different positions to capture the dependence between pixels. The model can therefore expand the multi-scale receptive field and have stronger context-sensitive ability, so as to improve the accuracy of semantic segmentation. Since the self-attention mechanism can focus on important details, the MBIE module can better understand and represent the details in the scene, and improve the result of semantic segmentation.

Feature Information Extraction Module

To perform further feature extraction from the three low-level outputs of the encoder effectively, a LBIE module is designed to improve the performance of feature mapping by collecting relevant semantic classes and eliminating redundant data. Moreover, self-attention is used to map the output features of each stage to provide an effective global context prior. Inspired by self-attention, the aim of the LBIE module is to weight and sum the context information based on the feature vector, simulate the long-distance dependence of the information, and capture the global correlation in the image. The architecture of LBIE is shown in Fig. 2.

$$\text{Self_attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

Considering the low-level features have higher spatial resolution and rich detail information than high-level features, they can better capture local details and textures. Therefore, by enhancing the semantic information of low-level features and simulating long-range dependencies to capture the global correlation to enable stronger global perception capabilities, it can help models better identify complex semantic boundaries and object shapes, thus improve the accuracy of semantic segmentation. Taking the input feature map F_2 as an example, the feature is extracted by using dilated convolution, which makes the network can process a wider range of context information and reduce redundant computation. Then the two identical parallel branches F_q , F_k are generated by 1×1 convolution

$$F_q = F_k = \text{Conv}_{1 \times 1}(w \odot F_2) \quad (7)$$

where w is the kernel of dilated convolution. When performing feature extraction on F_2 , the dilation rate is $d = 3$. When performing feature extraction on F_3 and F_4 , $d = 6, 9$, respectively. The dimensions are merged by reshape operation for F_q , and reshape and transpose operations are performed on F_k at the same time. Then, the inner product of the former two matrixes is performed and the channel weight δ are extracted by the activation function as

$$\delta = \sigma \{ \text{Res}(F_q) * \text{Per}[\text{Res}(F_k)] \} \quad (8)$$

where σ refers to the weight extracted with the Softmax activation function.

Finally, F_2 is fused with the extracted weights through 1×1 convolution, which is then directly added with the input feature F_2 to prevent the problem of gradient disappearance and gradient explosion. The output feature F_2^f of the LBIE module is obtained as

$$F_2^f = \text{Conv}_{1 \times 1}(F_2) \cdot \delta + F_2 \quad (9)$$

Feature Reuse Fusion Module

To keep the spatial and semantic information simultaneously, the FRF module is designed to capture the boundary information, and further perform context aggregation. The spatial resolution of the deep features is relatively low and some detail information is lost after multiple down-sampling. To restore fine-grained pixel-level semantic information, the low-level features are fused with the high-level features of the decoder. By learning the aggregated features from different levels, boundary features are combined with

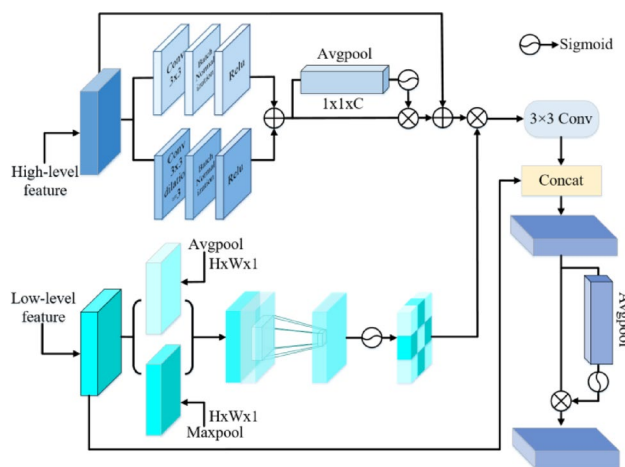


Fig. 3 The architecture of the FRF module

semantic feature maps. Feature reuse is used to share feature representation among different levels or modules of the network, allowing the network to capture different features of objects at different scales, thus improving the performance of semantic segmentation. The architecture of the feature reuse module is shown in Fig. 3.

First, the low-level and high-level features extracted by detail information are further used for weight. For the high-level feature F_H , a 3×3 convolution and a dilated convolution with $d = 3$ are used for multi-scale feature fusion to obtain F_H^1 , which is pooled in the spatial dimension, and the semantic weight $\delta_c \in R^{C \times 1 \times 1}$ is extracted through channel attention

$$\delta_c = \varphi [\text{Avg}_c(F_H^1)] \quad (10)$$

where φ refers to the sigmoid activation function, and Avg_c refers to average pooling in spatial dimension. The semantic weight δ_c is used to improve semantic feature representation of F_H^1 , which is calibrated and added to the feature map of F_H to obtain the high-level feature F_H^δ which is rich in semantic information

$$F_H^\delta = \delta_c \cdot F_H^1 + F_H \quad (11)$$

Second, average and max pooling operations are conducted on the low-level feature F_L on the channel dimension, respectively. The outputs with two channels of 1 are merged together through a 7×7 convolution, and the sigmoid activation function is used to extract the spatial weight δ_s

$$\delta_s = \varphi \{ \text{Conv}_{7 \times 7} [\text{Avg}_s(F_L) + \text{Max}_s(F_L)] \} \quad (12)$$

where Avg_s , Max_s refers to average pooling and max pooling in the channel dimension, respectively.

Third, F_H^δ is weighted by the spatial weight δ_s , and then the fusion output θ is obtained by combining 3×3 convolution and lower-level features. Finally, the semantic information is refined. Avg_c is applied to perform semantic extraction on the feature fusion output θ to obtain the final output F_f of the FRF module

$$\theta = \text{Cat}^{3 \times 3} [\text{Conv}_{3 \times 3} (\delta_s \cdot F_H^\delta) + F_L] \quad (13)$$

$$F_f = \varphi [\text{Avg}_c(F_H^1)] + \theta \quad (14)$$

The FRF module has the following advantages.

- (1) Making rational use of context information: a larger range of context information can be extracted by fusing low-level features with high-level features and combining attention mechanism, the semantic information of the image is modeled. It helps to solve challenges such as blurring object boundaries or segmentation of small objects.
- (2) Ensuring the accuracy of semantic information: features of different scales can be captured through feature reuse, which makes the semantic understanding of the target more accurate. By combining low-level and high-level features, the network can better classify different objects and obtain more accurate segmentation results.
- (3) Optimizing model parameters: Feature reuse enables networks to reduce the number of parameters by sharing parameters, thereby reducing the computational complexity of the model and reducing the storage space and time required for model training.

Loss Metric

The loss function plays a key role in the training process. The commonly used loss functions include the cross-entropy loss (CE Loss) [50] and the Dice loss (Dice Loss) [51].

CE Loss is used to measure the difference between the probability distribution predicted by the model and the ground truth. In semantic segmentation, the prediction of each pixel can be regarded as a classification problem, and each category corresponds to a channel. The CE loss calculates the cross-entropy between the predicted probability of each pixel position and the true label, and then sums or averages the loss of all pixel positions. By minimizing the CE loss, the model can learn more accurate pixel classification results.

$$L_C = -[Y_t \cdot \log(Y_P) + (1 - Y_t) \cdot \log(1 - Y_P)] \quad (15)$$

where L_C is the result of the cross-entropy loss, Y_t is the label of the sample. Y_P refers to the probability that the sample is predicted to be true.

Dice loss is used to evaluate the similarity between segmentation result and the real label. The Dice coefficient measures the degree of overlap between the predicted results and the real label. It can alleviate the problem of models that tend to predict categories with higher frequencies, and is therefore often used to deal with class imbalance situations

$$L_D = 1 - S = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \quad (16)$$

where L_D is the Dice Loss value, S is the Dice coefficient, and $X \cap Y$ is the intersection of the samples X and Y .

Compared with CE loss, Dice loss is more sensitive to the segmentation result of small targets and can improve the segmentation precision of small targets. Since the consistency at the regional level is more important than the accuracy at the pixel level in the road scene data set, when Dice Loss is 1.4 times, the model pays more attention to optimizing the segmentation effect at the regional level, thus improving the overall segmentation quality. The weight of 1.4 for Dice loss was determined through empirical experiments on the CamVid validation set, where it was observed to provide the best trade-off between pixel-level accuracy and region-level consistency, especially for class-imbalanced datasets. To balance pixel-level accuracy and region-level consistency, a mixed loss L_H is introduced in this work

$$L_H = L_C + 1.4 \cdot L_D \quad (17)$$

Experiments

To evaluate the performance of the proposed PBINet, experiments are conducted on CamVid and Cityscapes.

Datasets

CamVid is a collection of 701 high-resolution street view images from Cambridge, UK. The dataset covers urban scenarios with different weather conditions, times, and locations, includes 11 common urban scene categories, including Road, Symbol, Car, Sky, Sidewalk, Pole, Fence, Pedestrian, Building, Bicyclist, and Tree. Each image is provided with a pixel-level label. The initial resolution of the images is 720×960 . The dataset is divided into training set, verification set and test set, with 367,101 and 233 samples, respectively.

Cityscapes is widely used for semantic segmentation tasks, which contains street view images from German and other European cities, including 5,000 fine-annotated images and 20,000 coarse-annotated images. The initial resolution of the image is 1024×2048 . The dataset covers 19 semantic categories. The training set, validation set, and test set contain 2975, 500, and 1525 images, respectively. Since Cityscapes does not provide labels for test images, experiments are often performed on validation sets for ease of evaluation. To validate the real-time performance of the proposed network and to speed up the training process, the resolution of the images is adjusted to 512×1024 .

Evaluation Index

To evaluate the performance comprehensively, evaluation indexes including Pixel Accuracy (PA), Mean Pixel Accuracy (MPA), Intersection over Union (IoU), training parameter (M), detection time (T), and Mean Intersection over Union (mIoU) are used.

Pixel accuracy measures the ratio of the number of pixels that the model correctly classifies at the pixel level to the total number of pixels.

$$PA = \frac{\sum_{i=0}^k p_{ii}}{\sum_{i=0}^k \sum_{j=0}^k p_{ij}} \quad (18)$$

where i refers to the true value, j refers to the predicted value, and $(k+1)$ refers to the number of categories. Because there are differences in the number of pixels in different categories in the segmentation task, PA are unable to distinguish between different categories and are often evaluated in combination with mIoU.

MPA is used to calculate the ratio of the number of correctly classified pixels in each class, and then average all classes.

$$MPA = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij}} \quad (19)$$

The IoU is used to measure the degree of overlap between the predicted segmentation results and the real label by calculating the ratio of the intersection area and the union area between the predicted segmentation results and the real label.

The parameter quantity (Par) indicates the number of parameters to be learned in the model. The detection time (T) refers to the time spent in detecting a single image.

The mIoU measures the average accuracy of the segmented network in detecting each category by averaging the IoU for each category, with values ranging from 0 to 1.

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{p_{ii}}{\sum_{j=0}^k p_{ij} + \sum_{j=0}^k p_{ji} - p_{ii}} \quad (20)$$

Implementation Details

The ablation and comparative experiments are both trained and tested with 32G Tesla V100, python 3.7.0, pytorch 1.8.0, cuda 11.0, and cudnn. Adaptive Moment Estimation Optimizer (Adam) is used during the training process. The initial learning rate is set to $5e-4$, and the momentum and weight attenuation coefficients are set to 0.9 and 0, respectively. The batch size is set to 4 for all datasets. The learning strategy adopts cosine annealing attenuation (cos) strategy and adjusts the learning rate by cosine cycle

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left[1 + \cos \left(\frac{T_{cur}}{T_{max}} \pi \right) \right] \quad (21)$$

where η_t is the learning rate, η_{min} is the minimum learning rate, η_{max} is the initial learning rate, T_{cur} is the number of current training iterations, T_{max} is the number of iterations needed to reduce the learning rate to the minimum. The backbone network is MobileNetV2. To ensure the fairness of the comparative experiment, no data augmentation is used, and the pre-trained weight is used for the backbone, the number of iterations is 120 epochs. The resolution of the input of the CamVid dataset is 720×960 and that of the Cityscapes dataset is 512×1024 . The batch size is uniformly set to 4, and the network parameters are evaluated using the pytorch-torchstat library.

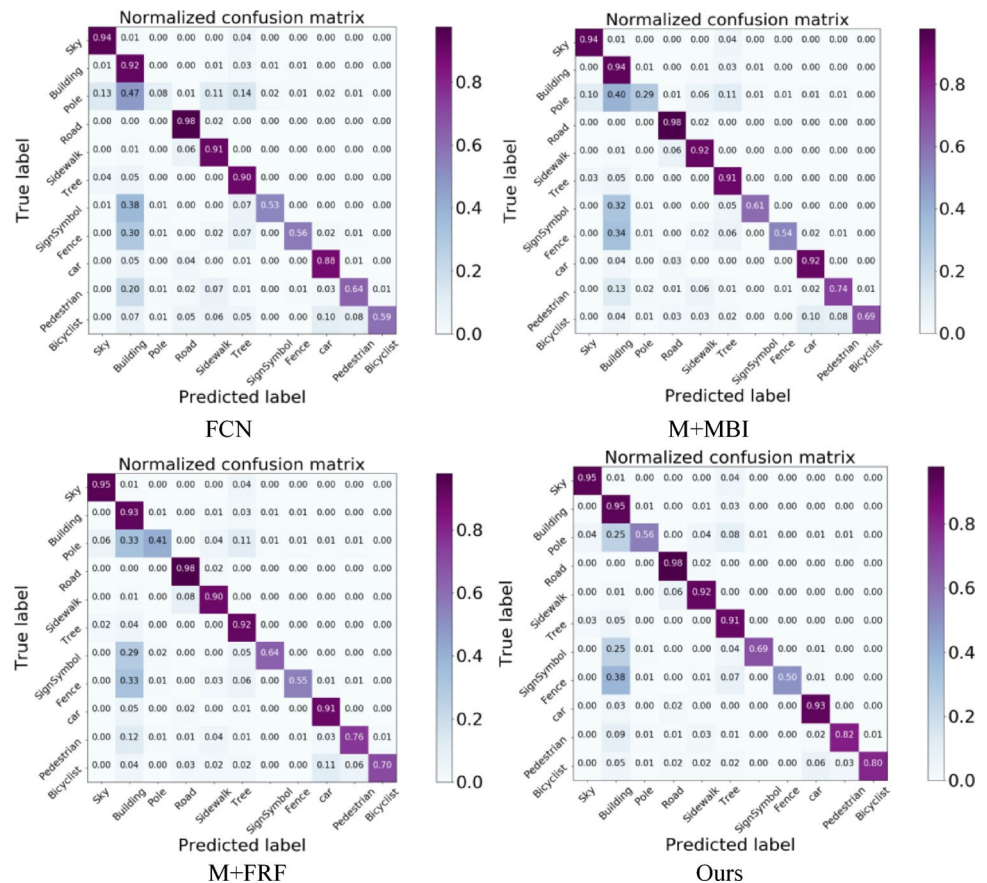
Ablation Experiment

To verify the validity of the modules proposed in PBINet, ablation experiments are conducted on the CamVid dataset. The FCN network with MobileNetV2 as the backbone is used as the baseline. As shown in Table 1, the baseline achieved a mIoU of 63.72%. After introducing the MBIE

Table 1 The ablation experiment of the proposed modules on CamVid dataset

Method	MBIE	FRF	LBIE	ML loss	Par(M)	mIoU
FCN					1.85	63.72
M+MBIE	√				2.14	69.43
M+FRF		√			2.28	70.61
M+MBIE+FRF	√	√			2.57	72.01
M+MBIE+FRF+LBIE	√	√	√		2.68	73.97
Ours	√	√	√	√	2.68	74.61

M represents the backbone network MobileNetV2

Fig. 4 Comparison of the precision of the confusion matrix**Table 2** Ablation experiments on the architecture of the MBIE module

$R = (5,9,13)$	$(6,12,18,24)$	$(1,5,9,13)$	CS	SA	Par(M)	mIoU
✓					5.25	72.08
	✓				6.22	72.43
		✓			6.22	73.07
		✓	✓		2.63	72.98
				✓	2.41	72.20
		✓	✓	✓	2.68	74.61

module, the mIoU increases from 63.72% to 69.43%. When the FRF module the residual structure of the decoder are applied to the backbone and, the mIoU increases from 63.72% to 70.61%. When the both modules are applied the mIoU increases to 72.01%. Further, after the LBIE module is introduced the mIoU increases by 1.96% to 73.97%. Finally, the mIoU of the proposed PBINet with ML loss reaches 74.61%, which is 10.89% higher than the baseline.

To visualize the detection accuracy of the ablation experiments for each category, the confusion matrix are shown in Fig. 4. It is shown that the detection accuracy of the proposed PBINet for each class is the highest, which validates the effectiveness of the proposed modules and the ML loss.

Table 3 The ablation experimental results of the LBIE module with three different feature extraction methods

Method	(1)	(2)	(3)	Ours
mIoU	72.33	72.85	73.70	74.61

The detailed ablation experiments are conducted for the architectures of the three proposed modules.

Ablation of MBIE

The MBIE module adopts a hybrid approach of self-attention and dilated convolution. To reduce computation, the MBIE module adopts channel grouping to reduce the network parameters significantly without affecting the accuracy. To

Table 4 The ablation experimental results on the architecture and the times of the application of the FRF module

Method	PA	MPA	Par	mIoU
(1)	92.93	79.17	2.30	72.13
(2)	92.98	79.33	2.48	72.53
(3)	93.18	79.37	2.68	72.50
(4)	93.08	79.79	2.68	72.84
F_{R1}	92.72	79.35	2.62	71.81
$F_{R1} + F_{R2}$	92.99	79.74	2.66	72.49
$F_{R1} + F_{R2} + F_{R3}$	93.41	81.85	2.68	74.61

prove the effectiveness, the contributions of self-attention (SA) and dilated convolution designed in this paper are verified respectively. As shown in Table 2, three combinations with different receptive fields, $R = (5,9,13)$, $(6,12,18,24)$, $(1,5,9,13)$, and whether to introduce channel shuffle (CS)

and SA, are evaluated. The experimental results prove the validity of each operation, and MBIE module can improve the segmentation accuracy while reducing the parameters.

Ablation of LBIE

For the LBIE module, the effect of cross-scale information cross-fusion in feature map and the necessity of extracting semantic information by using dilated convolution are discussed. The following experiments are conducted: (1) convolution is used to fuse information of different scales, (2) only attention is used for feature extraction, and (3) input information is fused after attention. The experimental results are shown in Table 3, the accuracy of the network with LBIE module is 2.28% higher than that of only dilated

Table 5 The comparative experimental results on the CamVid dataset with the state-of-the-art networks

Method	Backbone	PA	MPA	Par(m)	T(s)	Fps	mIoU
FCN [28]	Mobilenetv2	91.27	72.16	1.85	0.025	38.75	63.72
BiSeNet v1 [37]	Resnet18	92.61	77.38	12.75	0.031	32.6	69.17
LiteSeg [40]	Mobilenetv2	92.73	78.66	3.52	0.032	31.28	70.82
SwiftNet [38]	Resnet18	92.43	79.19	19.98	0.022	46.18	70.16
EMANet [34]	Resnet18	91.44	74.14	14.04	0.059	16.92	65.03
BAD [50]	Resnet18	92.81	77.66	11.73	0.035	28.83	70.53
Segformer [51]	Segformer b1	93.14	80.74	13.66	0.086	11.63	73.25
RGPNet [52]	Resnet18	93.06	79.89	18.19	0.025	40.63	71.71
JPANet [53]	Mobilenetv2	93.13	79.80	5.45	0.036	27.59	72.21
CSRNet [42]	Resnet18	91.62	76.02	11.55	0.044	22.94	71.28
Ours	Mobilenetv2	93.41	81.85	2.68	0.024	40.72	74.61

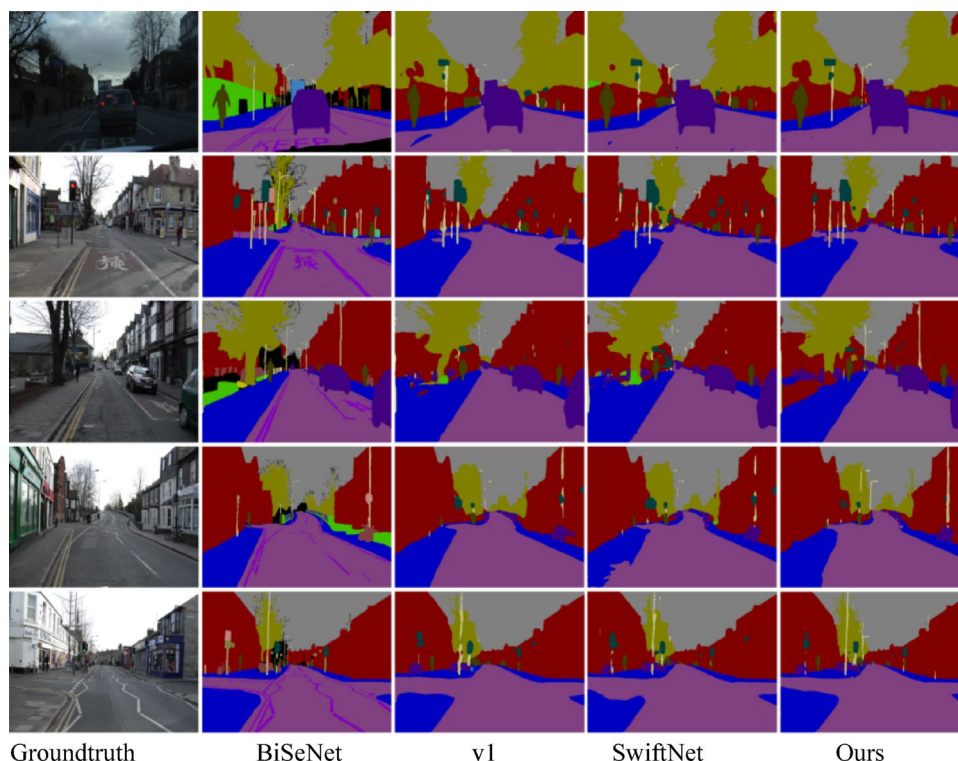
Fig. 5 The visualized segmentation results of the proposed method, BiSeNet and SwiftNet

Table 6 The comparative experimental results of each category on CamVid dataset

Methods	Sky	Bui	Pol	Roa	Sid	Tre	Sig	Fen	car	Ped	Bic	mIoU
FCN	89.8	83.3	7.1	94.6	82.7	77.1	41.6	45.0	81.9	47.2	50.4	63.72
BiSeNet v1	92.0	85.2	26.6	95.9	86.0	78.6	53.8	45.7	84.3	57.9	54.8	69.17
LiteSeg	92.7	85.7	35.8	95.0	84.1	79.1	53.6	42.4	87.5	62.7	60.3	70.82
SwiftNet	92.2	85.4	31.1	94.8	82.9	79.2	50.8	46.5	85.9	62.6	60.4	70.16
EMANet	89.7	83.1	8.8	95.2	84.1	76.7	41.8	43.8	83.7	49.1	89.1	65.03
BAD	92.5	85.7	29.6	95.6	85.4	78.6	53.1	49.0	86.7	62.2	57.4	70.53
Segformer	92.8	87.5	38.8	94.3	82.4	80.9	58.7	48.2	88.9	68.3	64.8	73.25
RGPNet	92.9	86.1	33.8	95.9	86.5	79.3	55.1	43.4	86.5	64.4	65.0	71.71
JPANet	92.6	86.4	37.6	95.7	85.9	79.5	55.7	44.1	86.7	65.8	64.2	72.21
CSRNet	90.8	82.9	36.1	94.9	83.8	77.3	50.1	34.2	83.2	61.7	45.4	70.28
Ours	92.6	87.3	43.2	95.6	85.9	79.9	59.4	47.2	88.7	68.8	72.3	74.61

convolution is used, and is 1.76% higher than that of only attentional feature extraction is used, and is 0.91% higher than that of input information after attentional mechanism, which validates the effectiveness of the proposed module.

Ablation of FRF

This section studies the ablation of the FRF module by discussing its internal architecture and the number of times it is used in decoder, and exploring the influence of context information on segmentation boundary refinement.

First, the internal architecture of the FRF module is studied, which is divided into the following four groups: (1) only a simple fusion is conducted of the two without processing; (2) do not perform dilated convolution on the input deep features to extract multi-scale information, nor average pooling in spatial dimensions to extract channel information; (3) do not perform spatial attention mechanism on the shallow input features to extract spatial information; and (4) cancel the semantic information refinement operation in the last part, and the feature output θ is taken as the final output directly.

Second, the times of the application of the FRF module is studied. The three FRF modules shown in Fig. 1 are labeled as F_{R1} , F_{R2} , and F_{R3} from bottom to top. Two additional groups of ablation experiments are conducted: the first group only used F_{R1} , and the second group used both F_{R1} and F_{R2} . The experimental results are shown in Table 4.

Comparison Study with the State-of-the-Arts

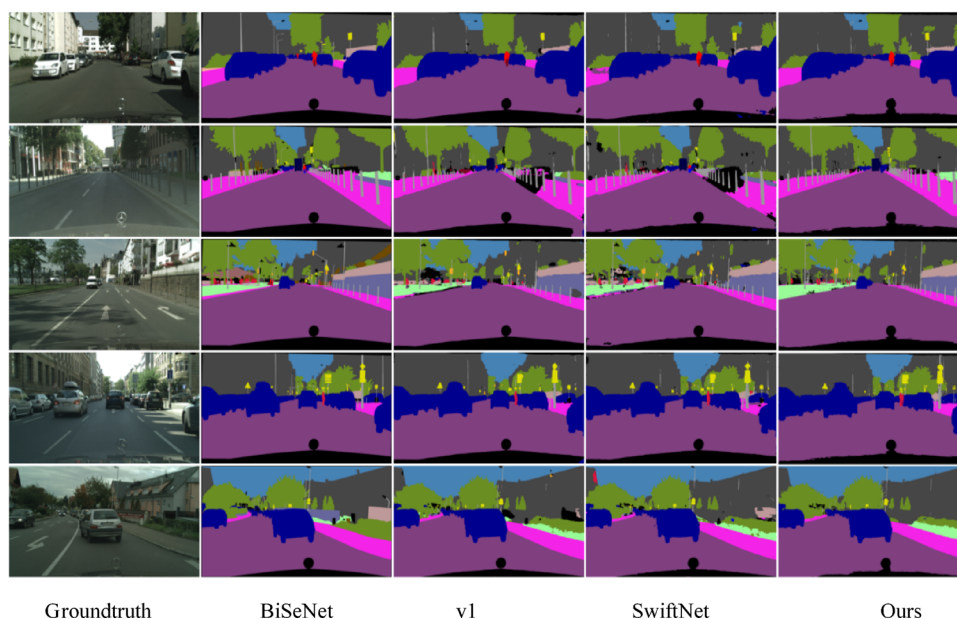
This section compares PBINet with the state-of-the-art semantic segmentation methods. Moreover, to show the advantage of lightweight network in parameter and precision, some non-lightweight semantic segmentation networks are selected for comparison. All networks load only the weight that pre-trained on ImageNet.

Experimental Results on CamVid

Table 5 shows the experimental results on the test set of CamVid. The state-of-the-art methods including FCN, BiSeNet V1, LiteSeg, SwiftNet, EMANet, BAD [52], Segformer [53], RGPNet [54], JPANet [55], and CSRNet are used for comparison. The proposed PBINet demonstrates superior performance over the current state-of-the-art networks. Specifically, it achieves improvements of 2.14% in PA, 9.69% in MPA, and 10.89% in mIoU compared to the FCN baseline. Compared with JPANet, the number of parameters of PBINet is two times lower while the mIoU is 2.40% higher. Table 6 shows the experimental results of the 11 classes. It is demonstrated that PBINet achieves mIoU

Table 7 The comparative experimental results on the Cityscapes dataset with the state-of-the-art networks

Method	Backbone	PA	MPA	Par(m)	T(s)	Fps	mIoU
FCN [28]	Mobilenetv2	88.04	66.36	1.85	0.059	16.87	55.61
BiSeNet v1 [37]	Resnet18	89.96	75.35	12.75	0.068	14.81	63.35
LiteSeg [40]	Mobilenetv2	90.46	76.45	3.52	0.057	17.53	64.45
SwiftNet [38]	Resnet18	90.21	76.69	19.98	0.057	17.68	65.1
EMANet [34]	Resnet18	88.5	71.14	14.04	0.092	10.92	59.08
BAD [50]	Resnet18	90.16	75.02	11.73	0.064	15.59	63.15
RGPNet [52]	Resnet18	90.64	77.53	18.19	0.058	17.10	66.44
JPANet [53]	Mobilenetv2	90.35	76.79	5.45	0.059	16.91	64.91
CSRNet [42]	Resnet18	90.23	76.31	11.55	0.058	17.10	67.44
Ours	Mobilenetv2	90.66	79.89	2.68	0.052	19.39	69.33

Fig. 6 The visualization of segmentation results on Cityscapes

improvements of 10.89% and 1.36% over FCN and Segformer, respectively. The above experimental results validate the effectiveness of the proposed PBINet on real-time performance and accuracy.

Figure 5 shows the visualized segmentation results of the proposed method, BiSeNet and SwiftNet. It is shown that the BiSeNet and SwiftNet cannot distinguish similar category boundaries in the image, e.g. the segmentation of sidewalk and Lane in Fig. 5. In comparison, the proposed PBINET can segment the edge of small objects, e.g. the telephone poles and street lights and signal lights, with satisfactory segmentation performance.

Experimental Results on Cityscapes

To further verify the generalization ability and robustness of the PBINet, comparative experiments are conducted on the

Cityscapes dataset, shown in Table 7. PBINet outperforms the baseline FCN network, achieving improvements of 2.62% in PA, 13.53% in MPA, and 13.72% in mIoU. It surpasses the top baseline CSRNet by 0.43% in PA, 3.58% in MPA, and achieves a 1.89% higher mIoU (69.33 vs. 67.44), while using only 23.2% of its parameters (2.68 M vs. 11.55 M). Compared with JPANet, has the number of the parameters of PBINet is two times lower while the mIoU is 4.42% higher, which proves its accuracy and efficiency.

Figure 6 shows the visualized predicted results of the proposed PBINet, BiSeNet V1 and SwiftNet. It is shown that the BiSeNet and SwiftNet cannot distinguish similar category boundaries (e.g. sidewalk vs. Greenbelt, car boundary in Fig. 6), while the PBINet has satisfactory segment performance on the edge of small objects (e.g. telephone poles, road signs, signs and pedestrians in the distance).

Conclusion

In this work, a pixel-wise boundary information refinement network PBINet is proposed for road scene segmentation, which the MBIE, LBIE and FRF modules are introduced. In the MBIE module, the channel splitting, atrous convolution, and self-attention are used to extract multi-scale boundary information in a lightweight manner. The LBIE module is used to refine the low-level outputs of the encoder. And the FRF module is then used to realize the effective fusion of the boundary information by efficiently using the low-level spatial information and the high-level semantic information. Moreover, the mixed loss function is designed to guide the network training. Experimental results on CamVid, and Cityscapes show that the proposed method can achieve satisfactory performance by virtue of making efficient use of the boundary information. Moreover, the proposed PBINet outperforms the state-of-the-art on semantic segmentation result with high precision and good real-time performance. Despite its strong performance, PBINet may struggle in extremely low-light conditions or with highly occluded objects, where boundary information is inherently ambiguous. Additionally, the model's reliance on boundary refinement may lead to over-segmentation in textured regions. Future work will focus on robustness to such challenging scenarios and further reducing computational overhead. Future directions include extending PBINet to video semantic segmentation by incorporating temporal consistency, exploring neural architecture search (NAS) for automatic module design, and adapting the model for multi-modal inputs (e.g., LiDAR and camera fusion) to enhance scene understanding in autonomous driving applications.

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Author Contributions Yunfeng Wang proposed the method and conducted the experiments, analyzed the data and wrote the manuscript. Yan Zhang provided ideas, evolution of overarching research goals and aims, critical review, project administration and funding acquisition. Others participated in comparative experiments investigation.

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Data Availability Data related to the current study are available from the corresponding author on reasonable request.

Code Availability Some of the codes generated or used during the study are available from the corresponding author by request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Li B, Shi Y, Qi Z, Chen Z. A survey on semantic segmentation. In: 2018 IEEE international conference on data mining workshops (ICDMW), IEEE pp. 1233–1240 (2018).
2. Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst.* 2012. <https://doi.org/10.1145/3065386>.
3. Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014).
4. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Rabinovich A. Going deeper with convolutions. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 1–9 (2015).
5. He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 770–778 (2016).
6. Chollet F. Xception: deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 1251–1258 (2017).
7. Hao S, Zhou Y, Guo Y. A brief survey on semantic segmentation with deep learning. *Neurocomputing.* 2020;406:302–21.
8. Zhao H, Shi J, Qi X, Wang X, Jia J. Pyramid scene parsing network. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 2881–2890 (2017).
9. Yu C, Wang J, Peng C, Gao C, Yu G, Sang N. Bisenet: bilateral segmentation network for real-time semantic segmentation. In: *Proceedings of the European conference on computer vision (ECCV)* pp. 325–341 (2018).
10. Nirkin Y, Wolf L, Hassner T. Hyperseg: patch-wise hypernetwork for real-time semantic segmentation. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* pp. 4061–4070 (2021).
11. Ronneberger O, Fischer P, Brox T. U-net: convolutional networks for biomedical image segmentation. In: *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III* 18 Springer International Publishing pp. 234–241 (2015).
12. Holder CJ, Shafique M. On efficient real-time semantic segmentation: a survey. *arXiv preprint arXiv:2206.08605*, (2022).
13. Papadeas I, Tsochatzidis L, Amanatiadis A, et al. Real-time semantic image segmentation with deep learning for autonomous driving: a survey. *Appl Sci.* 2021;11(19):8802.
14. Siam M, Gamal M, Abdel-Razek M, Yogamani S, Jagersand M, Zhang H. A comparative study of real-time semantic segmentation for autonomous driving. In: *Proceedings of the IEEE conference on computer vision and pattern recognition workshops* pp. 587–597 (2018).
15. Paszke A, Chaurasia A, Kim S et al. Enet: a deep neural network architecture for real-time semantic segmentation. *arXiv preprint arXiv:1606.02147*, (2016).
16. Sandler M, Howard A, Zhu M, Zhmoginov A, Chen LC. Mobilenetv2: inverted residuals and linear bottlenecks. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 4510–4520 (2018).
17. Zhang X, Zhou X, Lin M, Sun J. Shufflenet: an extremely efficient convolutional neural network for mobile devices. In: *Proceedings of the IEEE conference on computer vision and pattern recognition* pp. 6848–6856 (2018).
18. Tan M, Le Q. Efficientnet: rethinking model scaling for convolutional neural networks. In: *International conference on machine learning*. PMLR pp. 6105–6114 (2019).

19. Chollet F. Xception: deep learning with depthwise separable convolutions. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 1251–1258 (2017).
20. Mo Y, Wu Y, Yang X, et al. Review the state-of-the-art technologies of semantic segmentation based on deep learning. *Neurocomputing*. 2022;493:626–46.
21. Yu C, Wang J, Peng C et al. Bisenet: bilateral segmentation network for real-time semantic segmentation. In: Proceedings of the European conference on computer vision (ECCV) pp. 325–341 (2018).
22. Zhao H, Qi X, Shen X, Shi J, Jia J. ICNet for real-time semantic segmentation on high-resolution images. In: Proceedings of the European conference on computer vision (ECCV) pp. 405–420 (2018).
23. Poudel RPK, Liwicki S, Cipolla R. Fast-scnn: fast semantic segmentation network. *arXiv preprint arXiv:1902.04502*, (2019).
24. Li H, Xiong P, Fan H, Sun J. Dfanet: deep feature aggregation for real-time semantic segmentation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition pp. 9522–9531 (2019).
25. Sun K, Xiao B, Liu D, Wang J. Deep high-resolution representation learning for human pose estimation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition pp. 5693–5703 (2019).
26. Yu C, Xiao B, Gao C, Yuan L, Zhang L, Sang N, Wang J. Litehrnet: a lightweight high-resolution network. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition pp. 10440–10450 (2021).
27. Wang H, Jiang X, Ren H et al. Swiftnet: real-time video object segmentation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 1296–1305 (2021).
28. Brostow GJ, Shotton J, Fauqueur J et al. Segmentation and recognition using structure from motion point clouds. In: European conference on computer vision. Springer, Berlin, Heidelberg, pp. 44–57 (2008).
29. Cordts M, Omran M, Ramos S et al. The cityscapes dataset for semantic urban scene understanding. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 3213–3223 (2016).
30. Sohail A, Nawaz NA, Shah AA, et al. A systematic literature review on machine learning and deep learning methods for semantic segmentation. *IEEE Access*. 2022. <https://doi.org/10.1109/access.2022.3230983>.
31. Long J, Shelhamer E, Darrell T. Fully convolutional networks for semantic segmentation. In: Proceedings of the IEEE conference on computer vision and pattern recognition pp. 3431–3440 (2015).
32. Chen LC, Papandreou G, Kokkinos I et al. Semantic image segmentation with deep convolutional nets and fully connected crfs. *arXiv preprint arXiv:1412.7062*, (2014).
33. Yang M, Yu K, Zhang C, Li Z, Yang K. Denseaspp for semantic segmentation in street scenes. In: Proceedings of the IEEE conference on computer vision and pattern recognition pp. 3684–3692 (2018).
34. Fu J, Liu J, Tian H, Li Y, Bao Y, Fang Z, Lu H. Dual attention network for scene segmentation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition pp. 3146–3154 (2019).
35. Chen LC, Papandreou G, Kokkinos I, et al. Deeplab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Trans Pattern Anal Mach Intell*. 2017;40(4):834–48.
36. Chen LC, Zhu Y, Papandreou G, Schroff F, Adam H. Encoder-decoder with atrous separable convolution for semantic image segmentation. In: Proceedings of the European conference on computer vision (ECCV) pp. 801–818 (2018).
37. Li X, Zhong Z, Wu J, Yang Y, Lin Z, Liu H. Expectation-maximization attention networks for semantic segmentation. In: Proceedings of the IEEE/CVF international conference on computer vision pp. 9167–9176 (2019).
38. Badrinarayanan V, Kendall A, SegNet RC. A deep convolutional encoder-decoder architecture for image segmentation. *arXiv preprint arXiv:1511.00561*, (2015).
39. Mehta S, Rastegari M, Caspi A, Shapiro L, Hajishirzi H. Espnet: efficient spatial pyramid of dilated convolutions for semantic segmentation. In: Proceedings of the European conference on computer vision (ECCV) pp. 552–568 (2018).
40. Orsic M, Kreso I, Bevandic P, Segvic S. In defense of pre-trained imagenet architectures for real-time semantic segmentation of road-driving images. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition pp. 12607–12616 (2019).
41. Ma H, Yang H, Huang D. Boundary guided context aggregation for semantic segmentation. *arXiv preprint arXiv:2110.14587*, (2021).
42. Emarat T, Abd El Munim HE, Abbas HM, Liteseg. A novel lightweight convnet for semantic segmentation. In: 2019 digital image computing: techniques and applications (DICTA). IEEE, pp. 1–7 (2019).
43. Kumaar S, Lyu Y, Nex F, Yang MY. Cabinet: efficient context aggregation network for low-latency semantic segmentation. In: 2021 IEEE international conference on robotics and automation (ICRA). IEEE pp. 13517–13524 (2021).
44. Xiong J, Po LM, Yu WY, et al. CSRNNet: cascaded selective resolution network for real-time semantic segmentation. *Expert Syst Appl*. 2023;211:118537.
45. Hu J, Shen L, Sun G. Squeeze-and-excitation networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 7132–7141 (2018).
46. Woo S, Park J, Lee JY, Kweon IS. Cbam: convolutional block attention module. In: Proceedings of the European conference on computer vision (ECCV) pp. 3–19 (2018).
47. Huang S, Wang D, Wu X, Tang A. Dsanet: dual self-attention network for multivariate time series forecasting. In: Proceedings of the 28th ACM international conference on information and knowledge management pp. 2129–2132 (2019).
48. Liu J, Zhang F, Zhou Z, et al. Bfmnet: bilateral feature fusion network with multi-scale context aggregation for real-time semantic segmentation. *Neurocomputing*. 2023;521:27–40.
49. Ulku I, Akagündüz E. A survey on deep learning-based architectures for semantic segmentation on 2d images. *Appl Artif Intell*. 2022;36(1):2032924.
50. De Boer P-T, Kroese DP, Mannor S, Rubinstein RY. A tutorial on the cross-entropy method. *Ann Oper Res*. 2005;134:19–67.
51. Milletari F, Navab N, Ahmadi SA. V-net: fully convolutional neural networks for volumetric medical image segmentation. In: 2016 fourth international conference on 3D vision (3DV). IEEE, pp. 565–571 (2016).
52. Peng C, Tian T, Chen C, et al. Bilateral attention decoder: a lightweight decoder for real-time semantic segmentation. *Neural Netw*. 2021;137:188–99.
53. Xie E, Wang W, Yu Z, et al. SegFormer: simple and efficient design for semantic segmentation with transformers. *Adv Neural Inf Process Syst*. 2021;34:12077–90.

54. Arani E, Marzban S, Pata A, Zonooz B. Rgpnnet: a real-time general purpose semantic segmentation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision pp. 3009–3018 (2021).
55. Hu X, Jing L, Sehar U. Joint pyramid attention network for real-time semantic segmentation of urban scenes. *Appl Intell.* 2022;52(1):580–94.

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